

## Supplementary Materials

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## Supplementary Materials

**Reliability of Personality Measures****Overview**

Reliability was evaluated for both the full baseline PID-5 questionnaire (220 items) and the brief EMA-based PID-5 assessment (15 items, 3 per domain). For the EMA measures, which involve repeated assessments nested within persons, we employed multilevel reliability estimation following Lai (2021), which distinguishes between-person reliability (consistency of person-level means) from within-person reliability (consistency of occasion-level deviations).

**Baseline PID-5 Questionnaire**

The full PID-5 was administered at baseline (T1) using the standard 220-item version. Reliability was computed using Cronbach’s alpha and McDonald’s omega for each domain and for the total score. Results are reported for both the full sample and a cleaned sample excluding participants who failed attention check items (catch trials embedded at positions 68 and 161 in the questionnaire).

**Results.** All domains showed acceptable to excellent internal consistency (Table S1). Psychoticism exhibited the highest reliability ( $\alpha = .87$ ,  $\omega = .90$ ), consistent with its larger item pool (33 items). Antagonism showed the lowest reliability ( $\alpha = .67$ ,  $\omega = .82$ ), though McDonald’s omega—which accounts for item heterogeneity—indicated adequate composite reliability. The total PID-5 score demonstrated excellent reliability ( $\alpha = .96$ ,  $\omega = .98$ ). Excluding participants who failed attention checks produced slightly lower but comparable estimates, indicating that careless responding had minimal impact on scale properties in this sample.

Table S1

*Internal consistency of baseline PID-5 domains.*

| Domain               | Items | $\alpha$ (Full) | $\omega$ (Full) | $\alpha$ (Clean) | $\omega$ (Clean) |
|----------------------|-------|-----------------|-----------------|------------------|------------------|
| Negative Affectivity | 23    | 0.783           | 0.851           | 0.767            | 0.843            |
| Detachment           | 22    | 0.754           | 0.844           | 0.741            | 0.840            |
| Antagonism           | 21    | 0.670           | 0.819           | 0.627            | 0.796            |
| Disinhibition        | 22    | 0.747           | 0.843           | 0.725            | 0.824            |
| Psychoticism         | 33    | 0.870           | 0.898           | 0.864            | 0.892            |
| Total PID-5          | 220   | 0.965           | 0.979           | 0.964            | 0.977            |

**EMA-Based Brief PID-5**

The brief PID-5 administered via EMA comprised 15 items (3 per domain), selected based on factor loadings and domain representativeness from prior validation work (Bottesi et al., 2024). Because these items were assessed repeatedly across approximately 20 occasions per participant, standard single-level reliability estimates are inappropriate. Instead, we employed the multilevel composite reliability framework described by Lai (2021), which partitions variance into within-person and between-person components and computes separate reliability indices for each level.

**Multilevel Reliability Framework.** Following Lai (2021), we estimated three reliability indices:

- $\alpha_{2L}$  (**Two-level alpha**): Overall reliability of observed scores, pooling within- and between-person variance. Relevant when scores are used without distinguishing levels.
- $\alpha_B$  (**Between-person alpha**): Reliability of person-level means (aggregated across occasions). This is the relevant index when EMA scores are averaged to create a single trait estimate per person, as in our moderation analyses.

- $\alpha_W$  (**Within-person alpha**): Reliability of occasion-level deviations from each person’s mean. Relevant for detecting momentary fluctuations in personality states.

The same decomposition was applied using McDonald’s omega ( $\omega_{2L}$ ,  $\omega_B$ ,  $\omega_W$ ), which relaxes the assumption of tau-equivalence.

**Results.** The multilevel reliability analysis revealed a clear pattern (Table S2):

1. **Between-person reliability was high** ( $\alpha_B = .87$ ,  $\omega_B = .85$ ). This indicates that person-level trait estimates derived from aggregating across EMA occasions are measured with good precision. This is the critical index for our moderation analyses, which use person-level latent trait scores as predictors.
2. **Within-person reliability was adequate** ( $\alpha_W = .73$ ,  $\omega_W = .72$ ). This suggests that the brief scale can detect meaningful occasion-to-occasion fluctuations in personality states, though with more measurement error than at the between-person level. This is expected given the brevity of the scale (15 items total).
3. **Two-level reliability was good** ( $\alpha_{2L} = .83$ ,  $\omega_{2L} = .81$ ), reflecting the overall quality of observed scores when levels are pooled.

Table S2

*Multilevel reliability of EMA-based brief PID-5 (15 items).*

| Reliability Index         | Estimate | 95% CI Lower | 95% CI Upper |
|---------------------------|----------|--------------|--------------|
| $\alpha_{2L}$ (two-level) | 0.826    | 0.802        | 0.845        |
| $\alpha_B$ (between)      | 0.866    | 0.835        | 0.889        |
| $\alpha_W$ (within)       | 0.727    | 0.687        | 0.758        |
| $\omega_{2L}$ (two-level) | 0.814    | —            | —            |
| $\omega_B$ (between)      | 0.852    | —            | —            |
| $\omega_W$ (within)       | 0.724    | —            | —            |

**Interpretation for the Present Study.** The high between-person reliability ( $\alpha_B = .87$ ) supports the validity of using EMA-derived personality scores as person-level moderators of vocal stress responses. Because our moderation model estimates latent trait scores from the repeated EMA observations (see Statistical Models section), measurement error is explicitly modeled rather than ignored. The multilevel measurement model in our Stan implementation can be viewed as formalizing the reliability structure documented here: the latent trait  $\theta_{id}$  represents the “true” person-level standing on domain  $d$ , while the occasion-specific observations  $X_{nd}$  are treated as noisy indicators with residual variance  $\sigma_d^{\text{ema}}$  capturing within-person fluctuation and measurement error.

The adequate within-person reliability ( $\alpha_W = .73$ ) also suggests that the brief EMA measure could support analyses of state-level personality-voice covariation, though such analyses would require denser sampling than the present design to achieve adequate statistical power.

### Descriptive Statistics

Descriptive statistics are reported for  $N = 119$  female participants. Participants contributed on average 27 EMA assessments ( $SD = 4.16$ , range = 12–31).

### Design and Coverage

Table S3 summarizes the study design, including the number of participants, voice observations, and EMA assessments. All 119 participants contributed usable voice observations in each assessment period (Baseline, Pre-exam, Post-exam), consistent with the balanced, three-phase design.

Table S3

*Study design and data coverage.*

| $N$ | EMA $M$ | EMA $SD$ | EMA Range | Voice Obs. (per period) |
|-----|---------|----------|-----------|-------------------------|
| 119 | 27      | 4.16     | 12–31     | 119                     |

### Definition and psychological significance of the three principal acoustic features

Table S4 summarizes the definition, physiological basis, and psychological significance of the three principal acoustic features discussed in this literature.

### Voice Outcomes by Assessment Period

Tables S5 and S6 report descriptive statistics for the two primary voice outcomes—fundamental frequency (F0) and normalized noise energy (NNE)—across the three assessment periods.

Across periods, mean F0 increased from Baseline (191.1 Hz) to Pre-exam (194.8 Hz) and decreased slightly at Post-exam (192.6 Hz), consistent with anticipatory stress-related elevation in pitch followed by partial recovery. NNE showed more negative values

Table S4

*Principal Acoustic Features of Speech: Definitions and Psychological Significance*

| Feature                       | Definition                                                                   | Physiological basis                                                                       | Psychological significance                                                                                                                                                   |
|-------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fundamental frequency (F0)    | Rate of vocal fold vibration (Hz); acoustic correlate of perceived pitch     | Cricothyroid and vocalis muscle tension regulates vocal fold stiffness and vibratory rate | Most robust acoustic marker of stress: F0 increases reliably under acute stress, cognitive load, and evaluative threat via sympathetic activation                            |
| Normalized Noise Energy (NNE) | Ratio of inharmonic to harmonic spectral energy (dB); index of glottal noise | Degree of glottal closure completeness during phonation                                   | Reflects phonatory control and vocal effort; stress may increase noise (incomplete closure under arousal) or decrease it (compensatory hyperadduction and pressed phonation) |
| Jitter <sup>a</sup>           | Cycle-to-cycle variation in F0 period (%); measure of frequency perturbation | Irregularity of successive vocal fold vibratory cycles                                    | Proposed as an index of vocal instability and reduced neuromuscular control; elevated in some emotional and clinical states                                                  |

*Note.* F0 and NNE were retained as primary outcome variables in the present study.

<sup>a</sup>Jitter was extracted but not included in the main analyses for the following reasons:

(a) the literature on jitter and psychological stress is inconsistent, with effect sizes that are small, heterogeneous in direction, and often non-significant across studies (Giddens et al., 2013); (b) jitter is highly sensitive to recording conditions, signal-to-noise ratio, and extraction algorithm parameters, resulting in substantial measurement noise in field settings (Titze, 1994); and (c) unlike F0 and NNE, which index theoretically distinct mechanisms (autonomic arousal vs. phonatory control), jitter does not map onto a clearly separable psychophysiological pathway, limiting its interpretive value in the context of personality moderation hypotheses.

Table S5

*Voice outcome descriptives by assessment period.*

| Period    | <i>n</i> | F0 <i>M</i> (Hz) | F0 <i>SD</i> | NNE <i>M</i> (dB) | NNE <i>SD</i> |
|-----------|----------|------------------|--------------|-------------------|---------------|
| Baseline  | 119      | 191.11           | 21.30        | -26.43            | 2.63          |
| Pre-exam  | 119      | 194.81           | 21.36        | -27.12            | 3.25          |
| Post-exam | 119      | 192.63           | 23.12        | -26.93            | 2.87          |

(indicating reduced glottal noise) from Baseline to Pre-exam, with partial return toward Baseline in the Post-exam period.

Table S6 summarizes within-person variability across the three assessment periods, reporting the distribution of participant-specific standard deviations.

### **EMA Personality Domain Descriptives**

EMA-based PID-5 domain descriptives are reported separately for between-person variability (Table S7) and within-person variability (Table S8).

Between-person descriptives indicate substantial inter-individual variability in EMA trait levels, while within-person descriptives confirm meaningful intra-individual fluctuation across repeated assessments. This pattern is consistent with the multilevel measurement model used to estimate latent trait scores and propagate measurement uncertainty into moderation effects.



Table S6

*Within-person variability in voice parameters across periods.*

| Variable                    | $n$ | Mean $SD$ | $SD$ of $SD$ s | Min $SD$ | Max $SD$ |
|-----------------------------|-----|-----------|----------------|----------|----------|
| F0 within-person $SD$ (Hz)  | 119 | 7.44      | 5.47           | 0        | 28.76    |
| NNE within-person $SD$ (dB) | 119 | 1.73      | 1.11           | 0        | 5.58     |

### Statistical Models

#### Main Effects of Exam-Related Stress on Vocal Parameters

All analyses used hierarchical Bayesian models implemented in Stan via the `rstan` package, with four chains of 4,000 iterations each (2,000 warmup). Convergence was verified through R-hat statistics (all  $< 1.01$ ) and trace plot inspection. We report posterior medians with 95% credible intervals (CrIs) and directional probabilities.

**Research Question.** Before examining personality moderation, we first establish whether exam-related stress produces reliable changes in vocal acoustics. This model addresses the foundational question: Do vocal parameters (F0, NNE) change systematically across the three assessment phases—baseline, pre-exam (anticipatory stress), and post-exam (recovery)?

**Inferential Framework and Reporting.** All analyses were conducted within a Bayesian estimation framework, and we report and interpret full posterior distributions rather than dichotomous accept/reject decisions about individual parameters. Our inferential aim was to characterize the *direction*, *relative strength*, and *uncertainty* of the estimated associations and to interpret the overall pattern of results across parameters. For an exploratory study with a modest sample, this is the scientifically informative output of the analysis, whereas a binary verdict on each parameter would discard most of that information and invite the very significance-testing logic that a Bayesian treatment is meant to supersede (Kruschke & Liddell, 2018; McElreath, 2018).

Table S7

*EMA PID-5 descriptives: between-person variability (person means across occasions).*

| Domain               | $n$ | $M$  | $SD$ | Min  | Max  |
|----------------------|-----|------|------|------|------|
| Negative Affectivity | 119 | 4.53 | 1.62 | 0.71 | 8.08 |
| Detachment           | 119 | 2.04 | 1.62 | 0.00 | 6.19 |
| Antagonism           | 119 | 0.87 | 1.17 | 0.00 | 5.29 |
| Disinhibition        | 119 | 2.46 | 1.29 | 0.10 | 5.80 |
| Psychoticism         | 119 | 1.25 | 1.44 | 0.00 | 5.73 |

We summarize each parameter with three complementary quantities.

1. **Posterior median** — a point summary of the most probable parameter value given the model and the data.
2. **Probability of direction (pd)** — the posterior probability that the parameter takes its more probable sign,  $pd = \max[P(> 0), P(< 0)]$  (Makowski et al., 2019). The pd is our primary index of evidence. It ranges from .50 (the sign is completely uncertain) to 1.00 (the sign is effectively certain) and has a direct verbal reading: a pd of .97 means that, given the model and data, there is a 97% posterior probability that the effect lies in the stated direction. This is a probability statement *about the parameter*, which is available only in the Bayesian framework; under the frequentist framework the parameter is a fixed unknown and no probability can be attached to it. We treat the pd as a *continuous* index of directional evidence and deliberately impose no threshold above which an effect is declared “present”; we report it for every parameter and interpret the gradient of values.
3. **89% credible interval (equal-tailed)** — the central 89% of the posterior, reported as a summary of the *magnitude uncertainty* of the estimate. This interval is **not** a

Table S8

*EMA PID-5 descriptives: within-person variability (participant SDs across occasions).*

| Domain               | <i>n</i> | <i>M</i> of <i>SD</i> | <i>SD</i> of <i>SD</i> s | Min <i>SD</i> | Max <i>SD</i> |
|----------------------|----------|-----------------------|--------------------------|---------------|---------------|
| Negative Affectivity | 119      | 1.32                  | 0.42                     | 0.57          | 2.42          |
| Detachment           | 119      | 1.20                  | 0.62                     | 0.00          | 3.04          |
| Antagonism           | 119      | 0.71                  | 0.65                     | 0.00          | 2.84          |
| Disinhibition        | 119      | 1.17                  | 0.45                     | 0.30          | 2.31          |
| Psychoticism         | 119      | 0.93                  | 0.68                     | 0.00          | 2.93          |

significance test: whether or not it includes zero carries no special inferential status, and we do not interpret it that way. The 89% width (rather than 95%) is chosen deliberately and follows McElreath (2020): any interval width is arbitrary, and using a value other than 95% signals that the number is a descriptive convenience rather than a decision boundary, discouraging the reflexive mapping of “the 95% interval excludes zero” onto “ $p < .05$ .” A 50% interval, or any other width, would serve the same descriptive purpose; what matters is the location and spread of the posterior mass, not a cutoff.

Two consequences of this framework are worth making explicit. First, **we draw conclusions from the pattern of results across the full set of parameters, not from the status of any single estimate.** The claim that personality moderation of vocal stress responses is domain- and phase-specific rests on the *joint* configuration of directional probabilities across the ten F0 and ten NNE moderation parameters, not on any one interaction crossing a threshold. Second, **direction is better identified than magnitude.** As shown in the prior-sensitivity analysis (Supplement, *Prior-Sensitivity Analysis*), the sign and directional probability of the moderation effects are robust across prior specifications, whereas their point magnitudes are only weakly constrained by the

present sample. We therefore foreground direction and directional evidence (pd) and treat all reported magnitudes — and the widths of their credible intervals — as estimates carrying substantial uncertainty (Gelman & Carlin, 2014).

We did not use Bayes factors or region-of-practical-equivalence (ROPE) decision rules. Bayes factors reintroduce a thresholded, model-selection logic and are sensitive to the prior on the parameter under test — undesirable here, since the prior-sensitivity analysis shows the moderation magnitudes to be prior-dependent. A ROPE would require specifying a region of practical equivalence in the original vocal units (Hz, dB), for which there is no established basis in this novel measurement context. The probability of direction provides a more transparent and assumption-light summary of the evidence for our exploratory, directional questions.

**Model Specification.** The model is a standard Bayesian hierarchical (multilevel) linear regression with repeated measures nested within participants. Stress effects are parameterized using two orthogonal contrasts:

- **Stress contrast** ( $c_1$ ): Compares pre-exam to baseline, capturing anticipatory stress-induced change.
- **Recovery contrast** ( $c_2$ ): Compares post-exam to pre-exam, capturing post-stressor trajectory.

Let  $y_n$  denote the vocal outcome for observation  $n$ , with participant index  $s[n]$ . The model specifies:

$$y_n \sim \text{Normal}(\mu_n, \sigma_y)$$

where the linear predictor includes both fixed and random effects:

$$\mu_n = \alpha + u_{0,s} + (\beta_1 + u_{1,s}) \cdot c_{1n} + (\beta_2 + u_{2,s}) \cdot c_{2n}$$

The parameters are:

- $\alpha$ : Grand intercept (population-average baseline vocal level).
- $\beta_1$ : Population-average stress effect (pre-exam vs. baseline).
- $\beta_2$ : Population-average recovery effect (post-exam vs. pre-exam).
- $u_{0,s}$ : Participant-specific random intercept.
- $u_{1,s}$ : Participant-specific random slope for stress.
- $u_{2,s}$ : Participant-specific random slope for recovery.
- $\sigma_y$ : Residual standard deviation.

The random effects capture individual differences in baseline vocal characteristics ( $u_0$ ), stress reactivity ( $u_1$ ), and recovery patterns ( $u_2$ ). A non-centered parameterization is used for computational efficiency:

$$u_{k,s} = \tau_k \cdot z_{k,s}, \quad z_{k,s} \sim \text{Normal}(0, 1)$$

where  $\tau_k$  are the random effect standard deviations.

**Prior Specification.** Priors were chosen to be weakly informative, incorporating domain knowledge about plausible parameter ranges while allowing the data to dominate inference (Table S9).

**Interpretation.** The key parameters of interest are  $\beta_1$  (stress effect) and  $\beta_2$  (recovery effect):

- A positive  $\beta_1$  indicates that F0 increases from baseline to pre-exam, consistent with heightened autonomic arousal elevating vocal pitch.

Table S9

*Prior specifications for the main effects model.*

| Parameter          | Prior            | Rationale                                            |
|--------------------|------------------|------------------------------------------------------|
| $\alpha$           | Normal(220, 30)  | Centered on typical female F0 (Hz)                   |
| $\beta_1, \beta_2$ | Normal(0, 10)    | Weakly informative; allows effects up to $\pm 20$ Hz |
| $\tau_k$           | Exponential(0.5) | Weakly informative for random effect SDs             |
| $\sigma_y$         | Exponential(0.1) | Allows residual SD in plausible range                |

- A negative  $\beta_2$  would indicate that F0 decreases from pre-exam to post-exam, suggesting recovery toward baseline levels.

The random effect standard deviations ( $\tau_1, \tau_2, \tau_3$ ) quantify the degree of individual differences in baseline levels, stress reactivity, and recovery patterns. Large values of  $\tau_2$  or  $\tau_3$  would indicate substantial heterogeneity in how participants respond to stress—heterogeneity that might be explained by personality traits, motivating the moderation analyses presented in the next section.

The `generated quantities` block produces posterior predictive samples for model checking and pointwise log-likelihoods for model comparison via LOO-CV.

## Personality Moderation of Vocal Stress Responses

**Research Question.** The central question addressed by this model is whether the effect of exam-related stress on vocal acoustics (F0, NNE) is moderated by individual differences in personality pathology. Specifically, we ask: Do the five PID-5 domains—Negative Affectivity, Detachment, Antagonism, Disinhibition, and Psychoticism—differentially amplify or attenuate the stress-induced changes in vocal parameters during (a) anticipatory stress and (b) post-stressor recovery?

**The Challenge: Personality Traits from Intensive Longitudinal Data.** A key methodological challenge in this study concerns how personality traits are represented in the moderation analysis. Each participant completed approximately 20 EMA assessments over 2.5 months, providing repeated measures of each PID-5 domain. A naive approach would aggregate these observations into a single person-level mean for each domain and use these means as predictors in a standard multilevel regression. However, this approach discards valuable information and fails to account for measurement error: the observed person-means are noisy estimates of the true latent traits, and treating them as known quantities underestimates uncertainty in the moderation effects.

Our model addresses this problem through a *joint measurement-and-outcome model* that simultaneously estimates latent personality traits from the EMA data and their moderating influence on vocal stress responses. This integrated approach has three key advantages:

1. **Measurement error correction:** Rather than using observed means as fixed predictors, the model estimates each participant’s true latent trait score ( $\theta_{id}$ ) as a parameter, with appropriate uncertainty. This uncertainty propagates into the moderation estimates, yielding appropriately calibrated credible intervals.
2. **Borrowing strength across observations:** The repeated EMA measurements inform the latent trait estimates through a measurement model, allowing the model to distinguish stable trait variance from occasion-specific fluctuations.
3. **Coherent uncertainty quantification:** Because the latent traits and their effects are estimated jointly, the posterior distributions for moderation parameters ( $\gamma_1, \gamma_2$ ) fully reflect uncertainty about both the traits themselves and their influence on vocal outcomes.

**Model Specification.** The model comprises two interconnected components: a *measurement model* for the EMA-based personality assessments and an *outcome model* for the vocal parameters.

**Measurement Model (EMA).** Let  $X_{nd}$  denote the observed score for participant  $i$  on domain  $d$  at EMA occasion  $n$ . The measurement model specifies:

$$X_{nd} \sim \text{Normal}(\theta_{i[n],d}, \sigma_d^{\text{ema}})$$

where  $\theta_{id}$  is the latent true trait score for participant  $i$  on domain  $d$ , and  $\sigma_d^{\text{ema}}$  captures occasion-to-occasion variability (including both state fluctuations and measurement error). The latent traits are given standard normal priors:

$$\theta_{id} \sim \text{Normal}(0, 1)$$

This formulation treats each participant’s approximately 20 EMA observations as repeated noisy indicators of a stable underlying trait, with the model learning both the trait estimates and the amount of occasion-level variability for each domain.

**Outcome Model (Vocal Parameters).** Let  $y_j$  denote the vocal outcome (F0 or NNE) for observation  $j$ , with participant index  $i[j]$ . Stress effects are parameterized using two orthogonal contrasts:

- $c_1$ : Stress contrast (pre-exam vs. baseline);
- $c_2$ : Recovery contrast (post-exam vs. pre-exam).

The outcome model specifies:

$$y_j \sim \text{Normal}(\mu_j, \sigma_y)$$



where the linear predictor  $\mu_j$  includes fixed effects, random effects, and the crucial trait  $\times$  contrast interactions:

$$\mu_j = \alpha + u_{i,1} + (\beta_1 + u_{i,2}) \cdot c_{1j} + (\beta_2 + u_{i,3}) \cdot c_{2j} + \sum_{d=1}^5 [a_d \cdot \theta_{id} + \gamma_{1d} \cdot c_{1j} \cdot \theta_{id} + \gamma_{2d} \cdot c_{2j} \cdot \theta_{id}]$$

The parameters are:

- $\alpha$ : Grand mean (baseline vocal parameter);
- $\beta_1, \beta_2$ : Population-average stress and recovery effects;
- $u_{i,1}, u_{i,2}, u_{i,3}$ : Participant-specific random intercept and slopes;
- $a_d$ : Main effect of trait  $d$  on baseline vocal level;
- $\gamma_{1d}$ : **Stress moderation**—how trait  $d$  amplifies or attenuates the stress effect;
- $\gamma_{2d}$ : **Recovery moderation**—how trait  $d$  shapes post-stressor trajectory.

The moderation parameters  $\gamma_{1d}$  and  $\gamma_{2d}$  are the quantities of primary theoretical interest. A positive  $\gamma_{1d}$  indicates that higher levels of trait  $d$  are associated with larger stress-induced changes in the vocal parameter.

**Random Effects Structure.** Participant-level random effects are specified using a non-centered parameterization for computational efficiency:

$$u_{i,k} = z_{i,k} \cdot \tau_k, \quad z_{i,k} \sim \text{Normal}(0, 1)$$

where  $\tau_k$  are the random effect standard deviations. This structure allows for individual differences in baseline levels ( $u_{i,1}$ ), stress reactivity ( $u_{i,2}$ ), and recovery ( $u_{i,3}$ ) beyond what is explained by the PID-5 traits.

**Prior Specification.** Priors were chosen to be weakly informative, incorporating domain knowledge about plausible parameter ranges while allowing the data to dominate inference (Table S10).

Table S10

*Prior specifications for the moderation model.*

| Parameter                  | Prior            | Rationale                               |
|----------------------------|------------------|-----------------------------------------|
| $\alpha$                   | Normal(220, 30)  | Centered on typical female F0 (Hz)      |
| $\beta_1, \beta_2$         | Normal(0, 10)    | Allows stress effects up to $\pm 20$ Hz |
| $a_d$                      | Normal(0, 5)     | Modest trait effects on baseline        |
| $\gamma_{1d}, \gamma_{2d}$ | Normal(0, 3)     | Regularization toward zero              |
| $\tau_k$                   | Exponential(0.5) | Weakly informative for SDs              |
| $\sigma_y$                 | Exponential(0.1) | Allows residual SD up to $\sim 10$ Hz   |
| $\sigma_d^{\text{ema}}$    | Exponential(1)   | Weakly informative for EMA variability  |

The priors on the moderation parameters ( $\gamma_{1d}, \gamma_{2d}$ ) provide modest regularization, shrinking estimates toward zero in the absence of strong evidence. This helps guard against overfitting given the 10 moderation parameters (5 domains  $\times$  2 contrasts) being estimated.

**Stan Implementation.** The model was implemented in Stan. The complete code is available in the online repository accompanying this article.

**Interpretation of Key Parameters.** The model yields posterior distributions for 10 moderation parameters of primary interest:

- $\gamma_{1,\text{NegAff}}, \dots, \gamma_{1,\text{Psych}}$ : How each PID-5 domain moderates stress-induced vocal change (stress contrast  $\times$  trait).
- $\gamma_{2,\text{NegAff}}, \dots, \gamma_{2,\text{Psych}}$ : How each domain moderates recovery-phase vocal change (recovery contrast  $\times$  trait).

These parameters are expressed in the original units of the vocal outcome (Hz for F0, dB for NNE) per standard deviation of the latent trait. For example,  $\gamma_{1,\text{NegAff}} = 3.0$  would indicate that a one-SD increase in latent Negative Affectivity is associated with an

additional 3 Hz increase in F0 during the stress phase, beyond the population-average stress effect.

**Posterior Predictive Checks.** The `generated quantities` block produces posterior predictive samples ( $y_{\text{rep}}$ ) for model checking. These were used to verify that the model adequately captured the distributional properties of the observed vocal data, including means, variances, and the pattern of individual differences across assessment phases.

### Within-Person Temporal Covariation

We examined whether momentary fluctuations in personality states—assessed via EMA immediately before each voice recording—covaried with concurrent F0 levels. This tests whether trait-level moderation effects have within-person analogs: do individuals show higher F0 when they report elevated negative affectivity in the moment?

We compared hierarchical specifications with fixed effects (population-average slopes only) versus random slopes (allowing individual differences in within-person associations). Random slopes models substantially outperformed fixed-effects specifications ( $R^2 = 35\%$  vs  $2.5\%$ ; ELPD difference  $\approx 40$  points). However, with only three observations per person, posterior distributions for individual slopes were extremely wide—99.7% of participant  $\times$  domain combinations yielded credible intervals spanning zero.

This pattern reveals a critical distinction: the population shows clear evidence of slope heterogeneity (nonzero  $\sigma_\beta$  for several domains), but we cannot reliably identify which specific individuals have nonzero slopes. This reflects statistical power limitations rather than modeling failure. Denser temporal sampling would be required to resolve individual-level within-person dynamics.

## MCMC Convergence Diagnostics

### Estimation Procedure

All Bayesian models were estimated using Hamiltonian Monte Carlo (HMC) via Stan (Carpenter et al., 2017), accessed through the cmdstanr package in R. For the primary F0 moderation model, we ran 4 independent chains with 2,000 warmup iterations and 6,000 sampling iterations per chain, yielding 24,000 total post-warmup draws. Sampler adaptation parameters were set conservatively to ensure reliable exploration of the posterior: `adapt_delta = 0.99` and `max_treedepth = 15`. The same estimation procedure was applied to the NNE moderation model.

### Convergence Assessment

Convergence was assessed using standard diagnostics recommended for HMC estimation (Vehtari et al., 2021):

1. **Divergent transitions:** No divergent transitions were observed in either model, indicating that the sampler explored the posterior without encountering regions of high curvature that would compromise inference.
2. **Maximum treedepth:** No iterations exceeded the maximum treedepth, suggesting that the sampler did not require truncation of its trajectory length.
3. **Energy Bayesian Fraction of Missing Information (E-BFMI):** All chains showed E-BFMI values well above the recommended threshold of 0.2, indicating adequate exploration of the energy distribution.
4.  $\hat{R}$  (**R-hat**): The potential scale reduction factor was computed for all parameters. For the F0 model, all  $\hat{R}$  values for key parameters (fixed effects, moderation coefficients, variance components) were below 1.01, and no parameters across the full model exceeded 1.05. Table S11 reports  $\hat{R}$  values for key parameters.

5. **Effective Sample Size (ESS):** Both bulk-ESS and tail-ESS were computed for all parameters. Bulk-ESS, which reflects the efficiency of sampling for posterior means and medians, exceeded 4,000 for all key parameters. Tail-ESS, which reflects sampling efficiency for posterior quantiles, exceeded 3,000 for all key parameters. These values substantially exceed the minimum recommended threshold of 400 (Vehtari et al., 2021).

## Visual Diagnostics

Trace plots for all key parameters showed excellent mixing across chains, with no visible trends or differences between chains. Representative trace plots are shown in Figure S1.

Autocorrelation plots indicated low autocorrelation at lag 1 and negligible autocorrelation beyond lag 10 for all parameters, consistent with the high effective sample sizes reported above.

## Posterior Predictive Checks

Model adequacy was assessed using posterior predictive checks (Gelman et al., 2013). Figure S2 shows the posterior predictive density overlaid on the observed F0 distribution. The model accurately captured the location, spread, and shape of the observed data.

Additional posterior predictive checks confirmed that the model adequately recovered:

- The mean F0 across all observations (observed: 192.5 Hz; 95% PPC interval: [191.2, 193.8])
- The standard deviation of F0 (observed: 22.1 Hz; 95% PPC interval: [20.8, 23.4])
- The pattern of means across assessment periods (Baseline < Post-exam < Pre-exam)

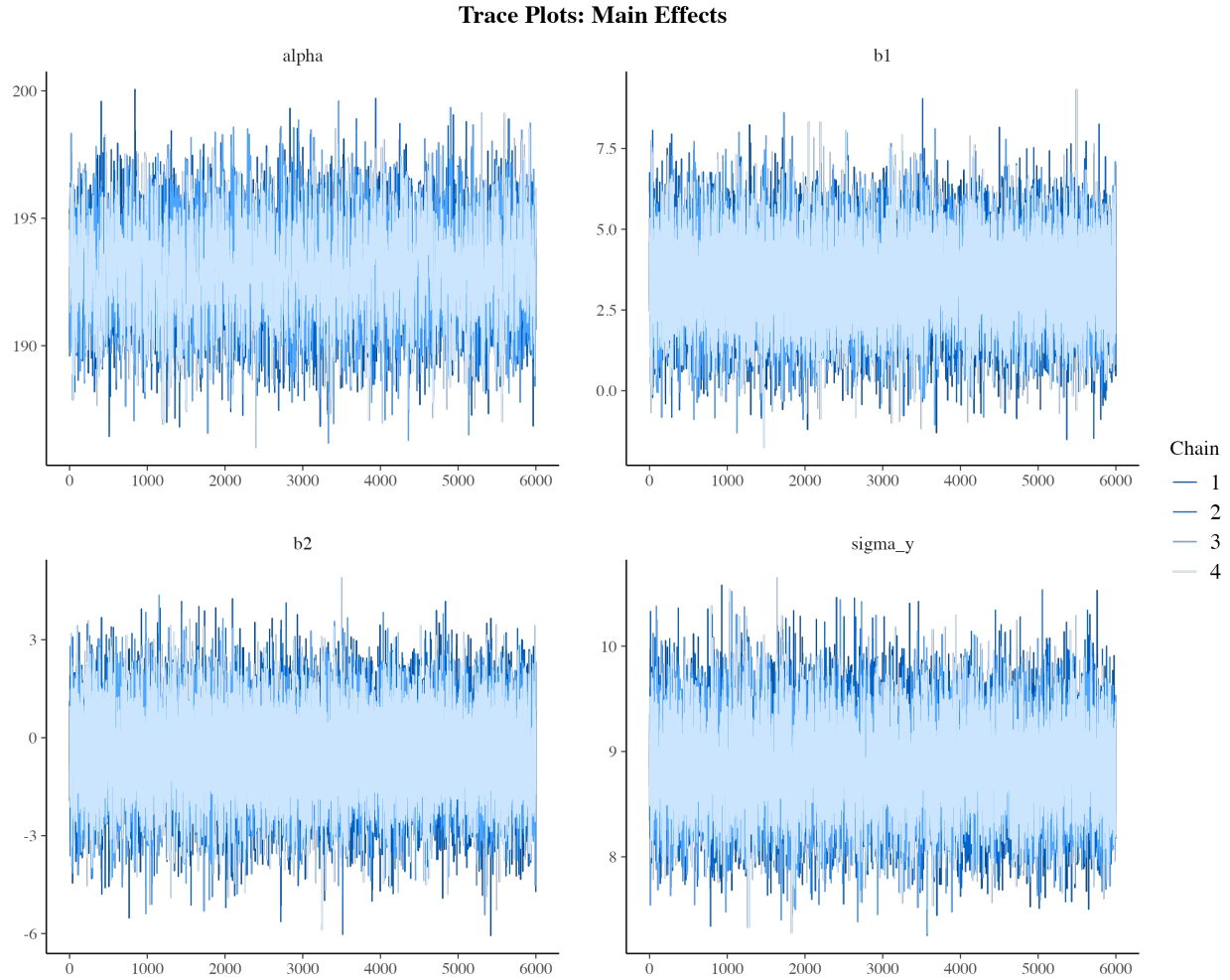
Table S11

*Posterior summary and convergence diagnostics for key parameters (F0 moderation model).*

| Parameter                             | Mean   | <i>SD</i> | $\hat{R}$ | ESS (bulk) | ESS (tail) |
|---------------------------------------|--------|-----------|-----------|------------|------------|
| Intercept ( $\alpha$ )                | 192.95 | 1.78      | 1.001     | 4490       | 8441       |
| Stress effect ( $\beta_1$ )           | 3.43   | 1.33      | 1.000     | 38288      | 19497      |
| Recovery effect ( $\beta_2$ )         | -0.49  | 1.34      | 1.000     | 38539      | 20141      |
| $\gamma_1$ Negative Affectivity       | 3.14   | 1.68      | 1.000     | 39739      | 20045      |
| $\gamma_1$ Detachment                 | -0.36  | 1.68      | 1.000     | 37929      | 20076      |
| $\gamma_1$ Antagonism                 | -0.10  | 1.61      | 1.000     | 38621      | 18464      |
| $\gamma_1$ Disinhibition              | -0.21  | 1.84      | 1.000     | 37799      | 19097      |
| $\gamma_1$ Psychoticism               | -0.26  | 1.71      | 1.000     | 38639      | 19968      |
| $\gamma_2$ Negative Affectivity       | -0.45  | 1.70      | 1.000     | 37632      | 19706      |
| $\gamma_2$ Detachment                 | -2.02  | 1.71      | 1.000     | 37665      | 19891      |
| $\gamma_2$ Antagonism                 | 3.16   | 1.61      | 1.000     | 37825      | 20302      |
| $\gamma_2$ Disinhibition              | -0.18  | 1.87      | 1.000     | 34125      | 19595      |
| $\gamma_2$ Psychoticism               | -1.42  | 1.70      | 1.000     | 39361      | 20089      |
| Residual SD ( $\sigma_y$ )            | 8.85   | 0.44      | 1.000     | 14363      | 17189      |
| Random intercept SD ( $\tau_1$ )      | 18.81  | 1.27      | 1.000     | 5855       | 11332      |
| Random stress slope SD ( $\tau_2$ )   | 1.33   | 1.14      | 1.000     | 12845      | 13041      |
| Random recovery slope SD ( $\tau_3$ ) | 1.65   | 1.43      | 1.000     | 10546      | 13300      |

## Summary

All convergence diagnostics indicated that the MCMC sampler successfully explored the posterior distribution. The absence of divergent transitions, combined with  $\hat{R}$  values below 1.01 and effective sample sizes exceeding 4,000, provides strong evidence that the posterior summaries reported in the main text are reliable. Posterior predictive checks



*Figure S1.* Trace plots for main fixed effects parameters showing good mixing across four chains.

confirmed that the model adequately captured the key features of the observed data.

## NNE Model Diagnostics

The same estimation and diagnostic procedures were applied to the NNE moderation model. Table S12 summarizes the convergence diagnostics for both models.



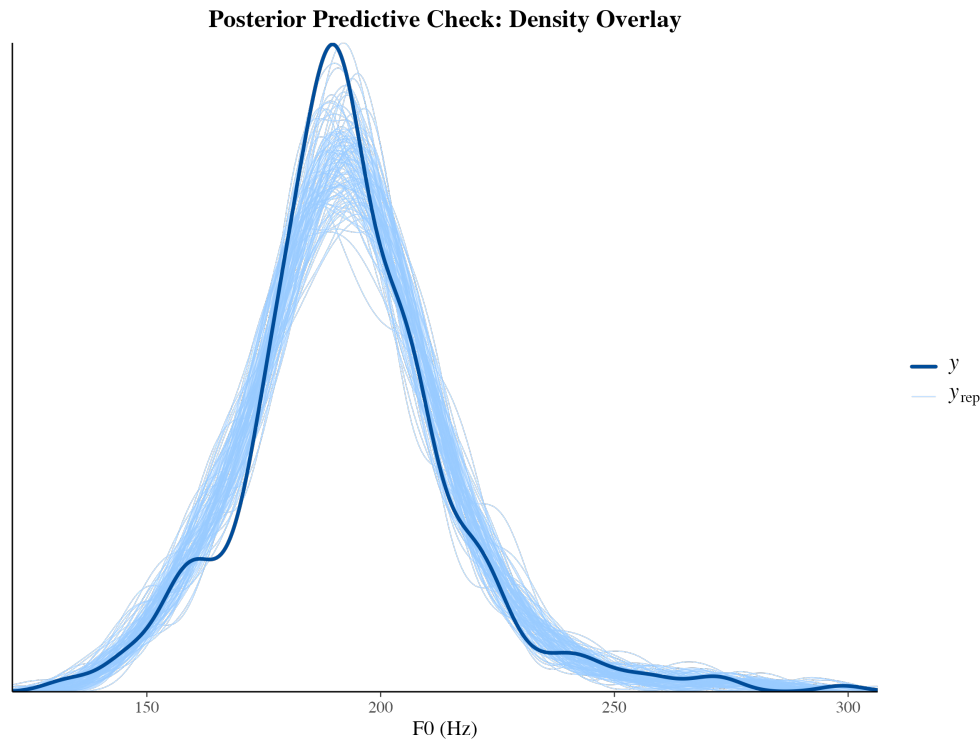


Figure S2. Posterior predictive check showing 100 draws from the posterior predictive distribution (light blue) overlaid on the observed F0 distribution (dark blue).

Table S12

Convergence diagnostics summary for F0 and NNE moderation models.

| Model | Divergences | Max Treedepth Exceeded | $\hat{R}$ (max) | $\hat{R} > 1.01$ | ESS Bulk (min) | ESS Tail (min) |
|-------|-------------|------------------------|-----------------|------------------|----------------|----------------|
| F0    | 0           | 0                      | 1.0015          | 0                | 4490           | 8441           |
| NNE   | 0           | 0                      | 1.0013          | 0                | 3775           | 4659           |

The NNE model showed equally good convergence properties: no divergent transitions, no iterations exceeding maximum treedepth, all  $\hat{R}$  values below 1.01, and minimum effective sample sizes well above recommended thresholds. Posterior predictive checks for the NNE model (Figure S3) confirmed adequate model fit.

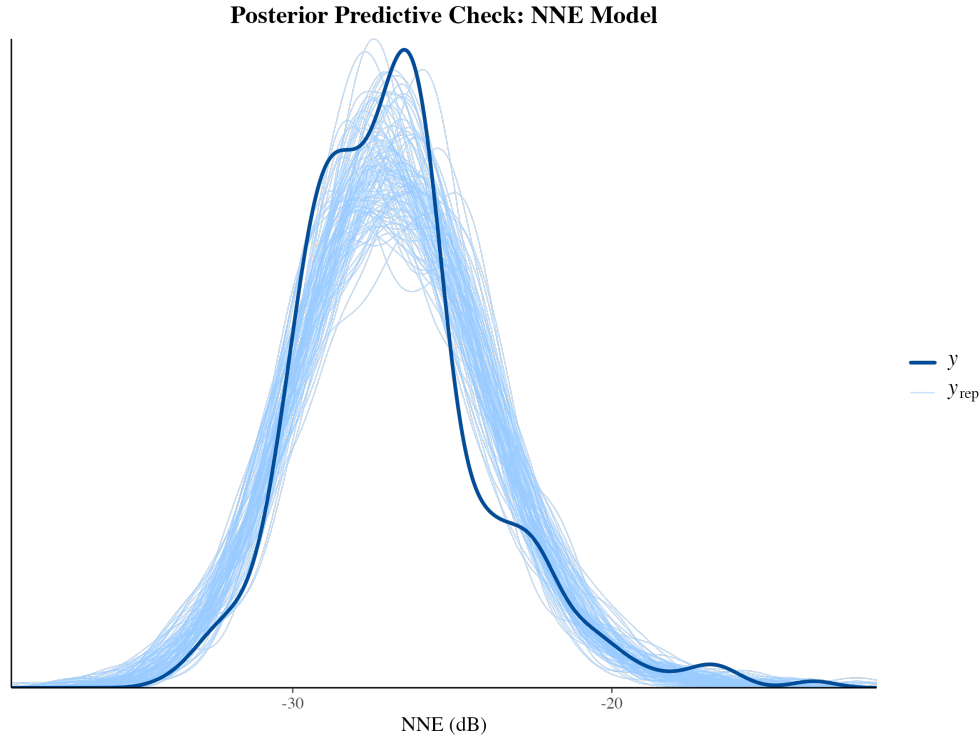


Figure S3. Posterior predictive check for the NNE moderation model.

### Posterior Predictive Assurance of Moderation Effects

#### Rationale

Beyond estimating posterior distributions for the moderation parameters, we conducted a posterior predictive assurance analysis to quantify the probability that the *direction* of the observed moderation effect would replicate under the same study design. Rather than addressing statistical significance or interval exclusion criteria, this analysis focuses on directional replicability: given the fitted model and the uncertainty in its parameters, how likely is it that a new study with the same design would yield a moderation effect in the same direction?

This approach is conceptually distinct from frequentist power analysis. Instead of assuming a fixed but unknown true effect size, posterior predictive assurance integrates over the full posterior distribution of the model parameters, thereby reflecting uncertainty

about both the magnitude of the effect and the data-generating process.

## Procedure

Posterior predictive simulations were conducted using draws from the joint posterior distribution of the fitted moderation model. For each simulation, a new dataset was generated under the same design as the original study, including the same number of participants, the same stress and recovery contrasts, and a comparable distribution of EMA observations per participant. Latent personality traits for the new participants were drawn from their population prior distributions, consistent with the generative assumptions of the model.

For each simulated dataset, the moderation effect of interest—specifically, the interaction between Negative Affectivity and the stress contrast—was re-estimated using a fast proxy model. Replication success was defined using a minimal and directionally focused criterion: the estimated moderation effect was required to be positive ( $> 0$ ). This criterion captures whether the effect would replicate in direction, without imposing additional thresholds related to statistical significance or effect size magnitude. This process was repeated 1,000 times, yielding an empirical estimate of the posterior predictive probability of directional replication (assurance).

## Results

Across posterior predictive replications, the moderation effect was positive in 92.1% of simulated datasets. The estimated posterior predictive probability of replication success was therefore 0.92, with a 95% credible interval of  $[0.90, 0.94]$ , reflecting Monte Carlo uncertainty in the simulation-based estimate.

## Interpretation

These results indicate a high probability that the direction of the moderation effect would replicate in a new sample drawn under the same design assumptions. In other words, given the fitted model and the uncertainty in its parameters, the interaction between Negative Affectivity and stress is expected to be positive in the large majority of replications.

At the same time, this analysis does not imply precise recovery of the effect magnitude. The posterior distribution of the moderation parameter remains relatively wide, indicating substantial uncertainty about the exact size of the effect. The assurance analysis therefore supports *directional robustness* of the moderation effect, while remaining agnostic about the degree of precision with which its magnitude can be estimated.

Taken together with the posterior estimates reported in the main analysis, these results suggest that the observed moderation effect is unlikely to be a chance reversal in direction, even though its quantitative strength should be interpreted with appropriate caution.

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